MSIN0304 | Data Management | Tools and Techniques

Assessment A2 | Group Report

Team 3 | UCL School of Management | 2025 to 2026

Clean Energy Investment Priority Index

A Data Product for ASEAN Country Prioritisation Under SDG 7

Prepared for

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1. SDG, Audience, and Decision Context

This project addresses SDG 7 (Affordable and Clean Energy), targeting both universal energy access (Target 7.1) and increased renewable share (Target 7.2) (United Nations, 2023). The analysis covers eight Southeast Asian economies, namely Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Thailand, and Vietnam, over the period 2015 to 2021. ASEAN represents a critical region where approximately 65 million people still lack electricity and over 200 million lack clean cooking fuels (IEA, 2023), yet FDI into the region's energy sector exceeded $30 billion in 2021 alone (World Bank, 2024a), presenting a clear convergence of investment opportunity and development need.

The data product is designed for Sarvesh Suri, IFC Regional Vice President for Asia and the Pacific, the senior decision maker allocating development finance across the region (IFC, 2024). It answers one specific capital allocation question, namely which ASEAN countries should receive IFC clean energy investment in the 2024 to 2030 window, balancing development impact with market readiness and economic viability. The answer is operationalised through a composite Clean Energy Investment Priority Index scored on a 0 to 100 scale, enabling direct cross country ranking and resource allocation.

2. Data Architecture and Integration

The pipeline follows a four stage lineage, beginning with raw external sources, passing through Python cleaning (00_clean_raw_data.py), then into SQL staging and integration (scripts 01 through 03), and finally into composite index calculation (05_priority_index.sql). Each stage is independently reproducible, and running the numbered scripts in sequence regenerates the final dataset from raw inputs. Full column level lineage is documented in data_dictionary.docx.

The integrated model uses a country by year grain, yielding 56 observations (8 countries across 7 years). A dimension table dim_country holds 8 records with ISO 3166 1 alpha 3 codes and income classifications. A fact spine of 56 rows, generated by a CROSS JOIN with a year sequence, establishes the complete analytical population before any source data is attached.

Four staging tables hold cleaned contributions from individual team members, covering IRENA and WDI renewable data (stg_renewable), IEA energy access snapshots (stg_access), WDI grid metrics (stg_grid), and IMF/World Bank economic indicators (stg_economic). The integration view v_integrated_master joins all four to the spine using LEFT JOIN on country_code and year, preserving the full 56 row population even where a staging table has gaps. The IEA access data provides only endpoint snapshots for 2015 and 2022, so stg_access joins on country_code alone and its values repeat across year rows. This is a structural feature of the source, not a cleaning artefact, and these columns are restricted to endpoint visuals in the dashboard. All monetary values use current USD, capacity uses MW, and percentages follow a 0 to 100 scale. The year ceiling of 2021 was set by the most temporally restrictive source (energy intensity), ensuring complete coverage across all dimensions.

3. Data Management Decisions and Quality Control

Country name standardisation. The eight sources used inconsistent naming. Lao PDR alone appeared as “Lao P.D.R.” in IMF data, “Laos” in IEA data, and “Lao People’s Democratic Republic (the)” in IRENA data. All variants were mapped to ISO alpha 3 codes before loading, ensuring every JOIN operates on a consistent key with no silent row loss.

IEA categorical value substitution. IEA reports categorical strings at distribution extremes, such as “>99%” and “<5%.” These were converted to numeric values (99, 3, and 0.5 respectively), and three boolean flag columns identify every substituted row, following IEA’s recommended practice (IEA, 2023).

Dual electricity access sources retained. WDI provides annual household survey data for trend analysis while IEA provides modelled endpoint estimates for gap comparison. The two diverge by up to 30.5 percentage points for Myanmar in 2015 (World Bank, 2024a; IEA, 2023), reflecting genuine methodological differences. Both are retained with source labelled column names to maintain traceability and avoid forcing a false reconciliation.

Composite index weighting justification. The Priority Index applies min max normalisation across all 56 rows. Three weighted components are computed, namely Impact Potential at 40%, Market Readiness at 30%, and Economic Viability at 30%. Impact Potential receives the highest weight because IFC’s mandate explicitly prioritises development impact over financial return (IFC, 2024), following the precedent of the UNDP Human Development Index, which assigns unequal dimension weights when institutional mandates warrant it (UNDP, 2022). To test robustness, the index was run under three alternative schemes, including equal weights (33/33/33), readiness led (25/40/35), and viability led (25/30/45). The Philippines and Indonesia remained in the top two across all configurations; however, Myanmar’s third place ranking proved sensitive to the impact led baseline, dropping to fourth under equal weights and below fifth when readiness or viability is emphasised. This confirms that the top two recommendation is robust while third tier allocation requires careful mandate alignment (see 05_priority_index.sql, QC checks 11 through 15).

Known anomalies. Five patterns were flagged and verified against original publications, including Malaysia’s near constant 6.9% T&D losses, Lao PDR’s sharp 2015 to 2016 grid improvement (Nam Theun 2 expansion), Vietnam’s 20,000 fold solar growth between 2015 and 2020 (feed in tariff; IRENA, 2024), Myanmar’s 12% GDP contraction in 2021 (military coup; IMF, 2024), and Thailand’s negative 2020 FDI (COVID 19 outflows; World Bank, 2024a). All were retained without modification.

Quality control. Fifteen structured checks across 04_qc_checks.sql and 05_priority_index.sql verified row count, completeness, null counts, year range, country count, five anomaly validations, and five index bound validations. All fifteen passed, and results are documented with screenshots in qc_results.pdf.

4. Data Product and Visual Communication

The Tableau dashboard is structured as a four page decision narrative designed to guide the IFC audience through sequential investment questions (Few, 2012), progressing from “where to prioritise” through to “where the greatest impact is achievable.”

The Priority Index Overview (Figure 1) shows which countries should be prioritised. A choropleth map and bar chart rank all eight countries by their composite index. Myanmar (mean 53.2) and the Philippines (47.6) emerge as top priority targets, followed by Indonesia (45.5). The dual encoding, combining geographic and ranked views, allows Suri to identify both spatial clusters and relative magnitudes at a glance.

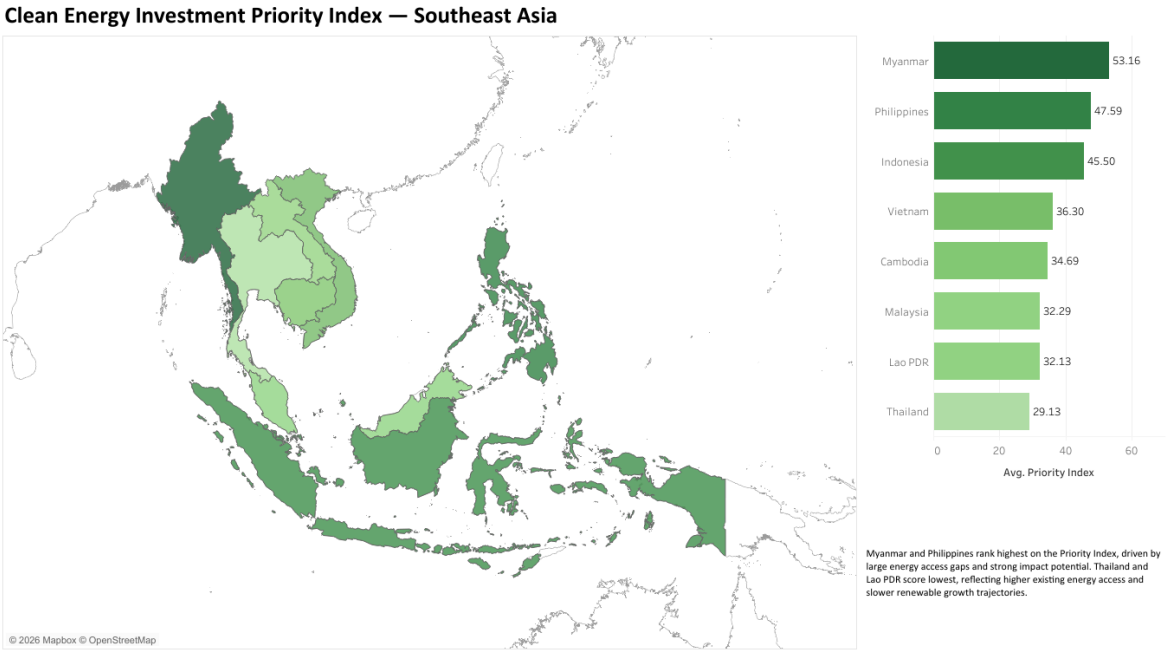


Figure 1. Priority Index Overview

Page 1 (Figure 2) shows where specifically to invest. A scatter plot maps GDP growth against renewable output with bubble size representing population, positioning Vietnam and Indonesia as the largest scale opportunities. Three KPI cards provide one actionable number per investment dimension. Vietnam's 16,656 MW solar capacity signals market maturity; Myanmar's 20 million people without electricity quantifies impact potential; and Indonesia's $21 billion FDI confirms absorptive capacity. A year slider (2015 to 2021) lets the audience trace trajectory shifts over time.

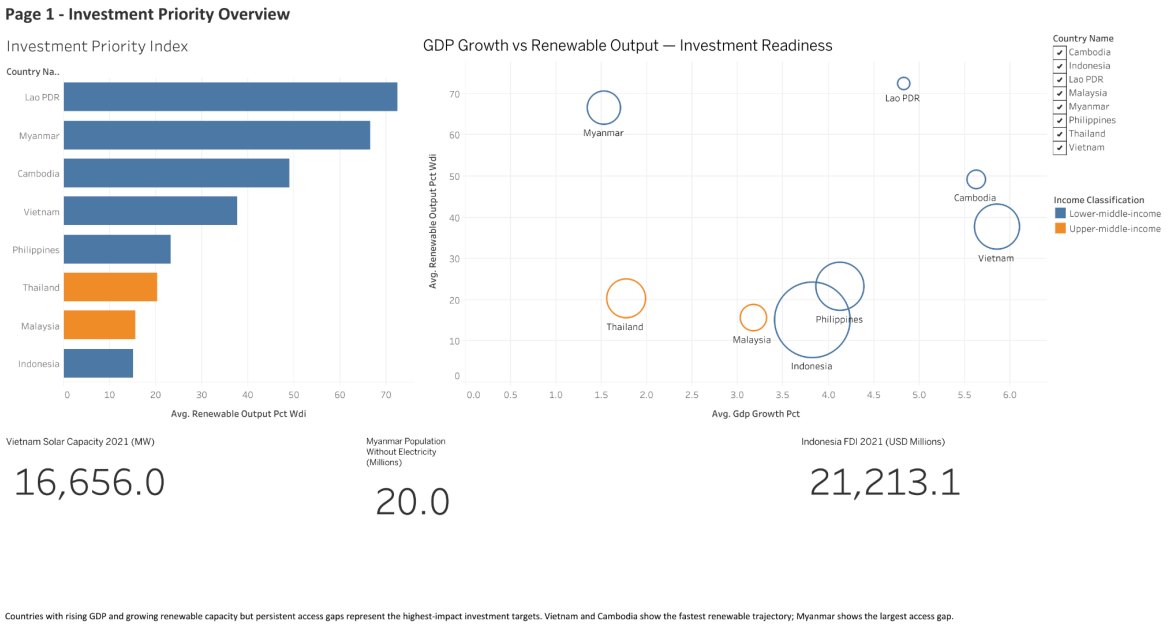


Figure 2. Investment Priority Overview

Page 2 (Figure 3) addresses when the grid is ready for investment. Line charts tracking T&D losses and renewable capacity reveal a sequencing pattern in which countries that reduced transmission losses earlier (Lao PDR between 2015 and 2016) saw steeper capacity increases by 2021, suggesting grid improvement as a leading indicator for investment timing. Vietnam's solar surge between 2019 and 2020, annotated as the feed in tariff effect, illustrates how policy signals precede capacity growth.

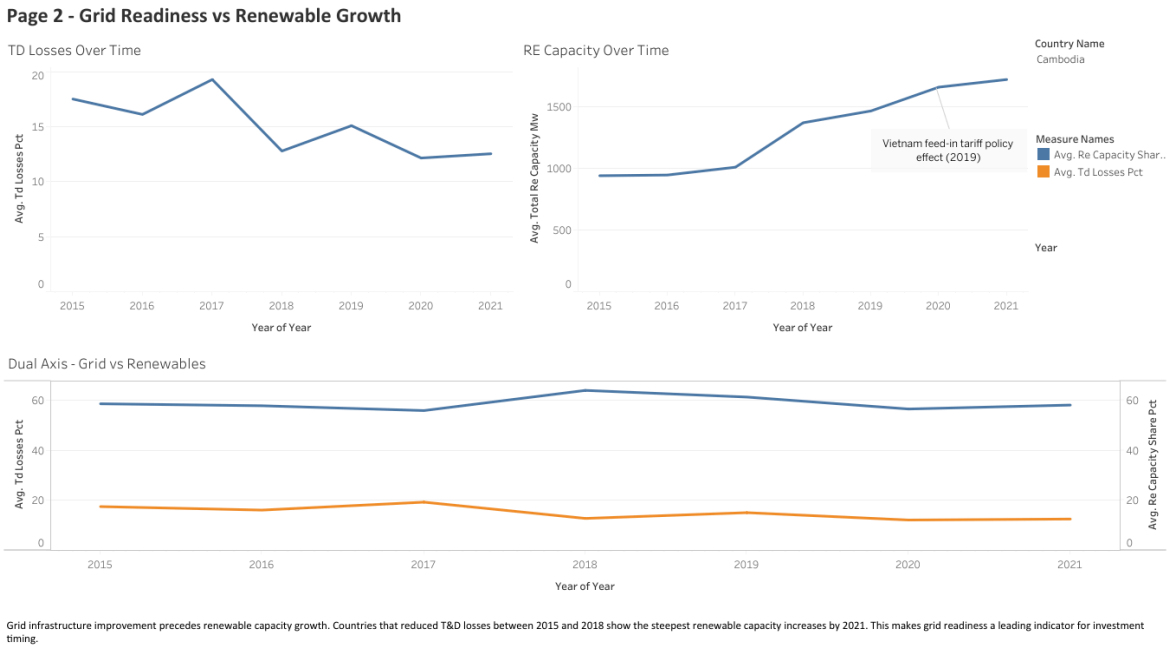


Figure 3. Grid Readiness vs Renewable Growth

Page 3 (Figure 4) addresses where the greatest impact is achievable. A connected dot plot comparing 2015 and 2022 electricity access rates shows Myanmar's growth from 30% to 64%, representing the largest remaining gap and fastest improvement. A bar chart of

population without clean cooking shows the Philippines at 60 million as the single largest impact opportunity. Urban versus rural bars highlight Myanmar's 50 percentage point divide as the most severe geographic inequality, directly relevant to IFC's sub national targeting.

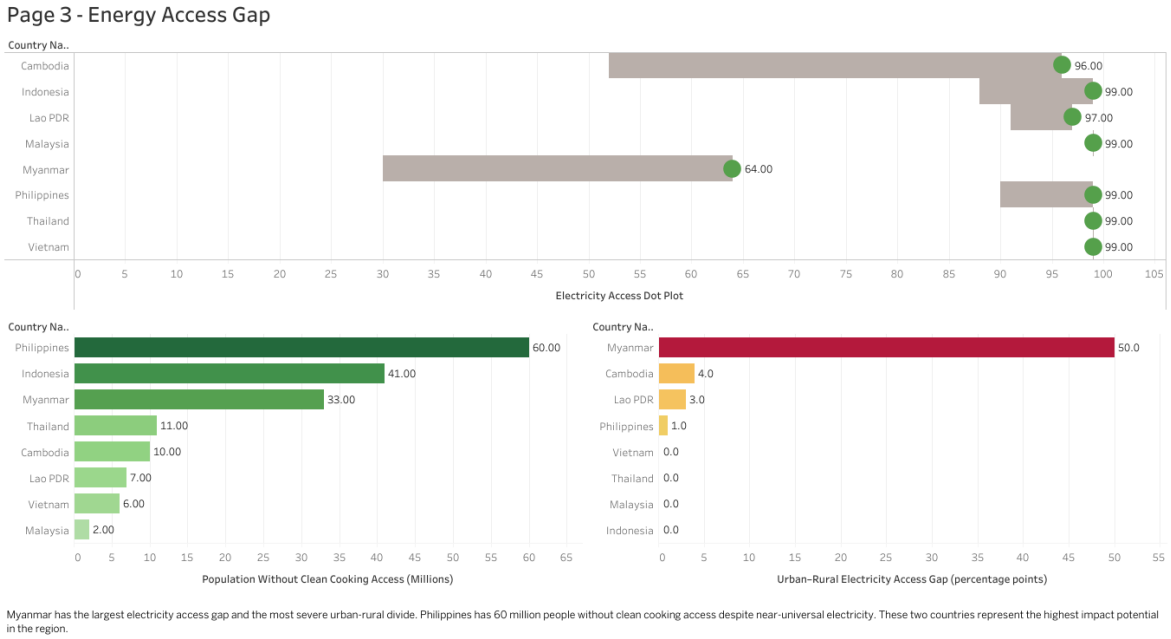


Figure 4. Energy Access Gap

5. Interpretation, Limitations, and Next Steps

The product supports country level prioritisation across four dimensions for eight ASEAN markets over 2015 to 2021. Under the baseline impact led weighting, Myanmar and the Philippines rank highest, combining large access gaps with strong improvement momentum, while Indonesia ranks third on the strength of its economic fundamentals. Sensitivity testing confirms that the Philippines and Indonesia hold the top two positions across all alternative weighting schemes, making them the most robust allocation targets, while Myanmar's ranking depends on the weight assigned to development impact relative to market readiness.

The product cannot support sub-national analysis, evaluate post 2021 policy changes, quantify political risk, or disaggregate grid connected from off grid electrification. The Impact Potential component relies on IEA snapshot data that does not vary annually, a structural limitation of the source rather than a modelling choice. Although the top two ranking proved robust, Myanmar's position is sensitive to the balance between impact and readiness. Incorporating political risk indicators and sub national grid data would strengthen discriminatory power in future iterations (Nardo et al., 2005).

6. References

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